

Assignment 1

Q. Configure iptables as a network firewall.

- Your network address is 192.168.25.0/24

```
root ~
❯ ip -br a
lo                UNKNOWN      127.0.0.1/8 ::1/128
ens160            UP          192.168.74.214/24 fe80::20c:29ff:feb4:c3a5/64
ens224            UP          192.168.25.254/24 fe80::92ef:a438:904f:de22/64
root ~
❯ route -n
Kernel IP routing table
Destination        Gateway            Genmask           Flags Metric Ref    Use Iface
0.0.0.0            192.168.74.254    0.0.0.0           UG        100    0      0 ens160
192.168.25.0       0.0.0.0           255.255.255.0     U         101    0      0 ens224
192.168.74.0       0.0.0.0           255.255.255.0     U         100    0      0 ens160
root ~
❯ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 192.168.72.20
```

Enable IP forwarding

```
root ~
❯ echo "--Enable IP forwarding--"
--Enable IP forwarding--
root ~
❯ cat /proc/sys/net/ipv4/ip_forward
1
root ~
❯ cat /etc/sysctl.conf | tail -2
#
net.ipv4.ip_forward = 1
```

- Configure 2 clients

Client M1:

```
[ndc2@localhost ~]$ ip -br a
lo                UNKNOWN          127.0.0.1/8 ::1/128
ens160            UP              192.168.25.10/24 fe80::da8d:816:efba:1563/64
[ndc2@localhost ~]$ route -n
Kernel IP routing table
Destination        Gateway            Genmask           Flags Metric Ref    Use Iface
0.0.0.0            192.168.25.254    0.0.0.0           UG    100    0      0 ens160
192.168.25.0       0.0.0.0           255.255.255.0     U     100    0      0 ens160
[ndc2@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 192.168.72.20
[ndc2@localhost ~]$ ping 192.168.25.254
PING 192.168.25.254 (192.168.25.254) 56(84) bytes of data.
64 bytes from 192.168.25.254: icmp_seq=1 ttl=64 time=0.584 ms
64 bytes from 192.168.25.254: icmp_seq=2 ttl=64 time=1.19 ms
^C
--- 192.168.25.254 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1019ms
rtt min/avg/max/mdev = 0.584/0.884/1.185/0.300 ms
[ndc2@localhost ~]$
```

Client M2:

```
[ndc3@localhost ~]$ ip -br a
lo                UNKNOWN          127.0.0.1/8 ::1/128
ens160            UP              192.168.25.20/24 fe80::f3c4:7227:d05c:568b/64
[ndc3@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 192.168.72.20
[ndc3@localhost ~]$ route -n
Kernel IP routing table
Destination        Gateway            Genmask           Flags Metric Ref    Use Iface
0.0.0.0            192.168.25.254    0.0.0.0           UG    100    0      0 ens160
192.168.25.0       0.0.0.0           255.255.255.0     U     100    0      0 ens160
[ndc3@localhost ~]$ ping 192.168.25.254
PING 192.168.25.254 (192.168.25.254) 56(84) bytes of data.
64 bytes from 192.168.25.254: icmp_seq=1 ttl=64 time=0.500 ms
64 bytes from 192.168.25.254: icmp_seq=2 ttl=64 time=1.11 ms
^C
--- 192.168.25.254 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1023ms
rtt min/avg/max/mdev = 0.500/0.806/1.112/0.306 ms
```

- Create a new chain named CDAC

```

root ~
🔗 echo "---Create new chain named CDAC---"
---Create new chain named CDAC---
root ~
🔗 iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination            state
ACCEPT     all  --  anywhere               anywhere               state RELATED,ESTABLISHED
ACCEPT     icmp --  anywhere               anywhere
ACCEPT     all  --  anywhere               anywhere
ACCEPT     tcp  --  anywhere               anywhere               state NEW tcp dpt:ssh
REJECT     all  --  anywhere               anywhere               reject-with icmp-host-prohib
ited

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
root ~
🔗 iptables -N CDAC
root ~
🔗 iptables -L CDAC
Chain CDAC (0 references)
target     prot opt source                destination

```

Allow DNS traffic

```

root ~
🔗 echo "---Allow DNS Queries---"
---Allow DNS Queries---
🔗 iptables -A CDAC -p udp -s 192.168.25.0/24 -d 192.168.72.20 --dport 53 -j ACCEPT
root ~
🔗 iptables -A CDAC -p udp -d 192.168.25.0/24 -s 192.168.72.20 --sport 53 -j ACCEPT
root ~
🔗 iptables -A CDAC -p tcp -s 192.168.25.0/24 -d 192.168.72.20 --dport 53 -j ACCEPT
root ~
🔗 iptables -A CDAC -p tcp -d 192.168.25.0/24 -s 192.168.72.20 --sport 53 -j ACCEPT
root ~
🔗 iptables -L CDAC
Chain CDAC (0 references)
target     prot opt source                destination
ACCEPT     udp  --  192.168.25.0/24        student-dc.students.acts.in  udp dpt:domain
ACCEPT     udp  --  student-dc.students.acts.in 192.168.25.0/24            udp spt:domain
ACCEPT     tcp  --  192.168.25.0/24        student-dc.students.acts.in  tcp dpt:domain
ACCEPT     tcp  --  student-dc.students.acts.in 192.168.25.0/24            tcp spt:domain
root ~
🔗 iptables -A CDAC -p tcp -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
root ~
🔗 iptables -L CDAC
Chain CDAC (0 references)
target     prot opt source                destination
ACCEPT     udp  --  192.168.25.0/24        student-dc.students.acts.in  udp dpt:domain
ACCEPT     udp  --  student-dc.students.acts.in 192.168.25.0/24            udp spt:domain
ACCEPT     tcp  --  192.168.25.0/24        student-dc.students.acts.in  tcp dpt:domain
ACCEPT     tcp  --  student-dc.students.acts.in 192.168.25.0/24            tcp spt:domain
ACCEPT     tcp  --  anywhere               anywhere                    ctstate RELATED,ESTABLISHED

```

Configure the CDAC chain such that:

Client 1: Allow access only to:

- Red Hat website
- Microsoft website
- Oracle website

```
root ~
# iptables -A CDAC -p tcp -s 192.168.25.10 -d www.redhat.com --dport 443 -m conntrack --ctstate NEW -j ACCEPT
root ~
# iptables -A CDAC -p tcp -s 192.168.25.10 -d www.microsoft.com --dport 443 -m conntrack --ctstate NEW -j ACCEPT
root ~
# iptables -A CDAC -p tcp -s 192.168.25.10 -d www.oracle.com --dport 443 -m conntrack --ctstate NEW -j ACCEPT
root ~
```

Client 2: Allow access only to:

- HDFC Bank website
- SBI Bank website

```
root ~
# iptables -A CDAC -p tcp -s 192.168.25.20 -d www.hdfc.bank.in --dport 443 -m conntrack --ctstate NEW -j ACCEPT
root ~
# iptables -A CDAC -p tcp -s 192.168.25.20 -d onlinesbi.sbi.bank.in --dport 443 -m conntrack --ctstate NEW -j ACCEPT
root ~
```

For All Clients: Allow access to:

- Google
- Website/IP: 192.168.72.20

```
root ~
# iptables -A CDAC -p tcp -s 192.168.25.0/24 -d www.google.com --dport 443 -m conntrack --ctstate NEW -j ACCEPT
root ~
# iptables -A CDAC -p tcp -s 192.168.25.0/24 -d 192.168.72.20 --dport 80 -m conntrack --ctstate NEW -j ACCEPT
root ~
# iptables -A CDAC -p tcp -s 192.168.25.0/24 -d 192.168.72.20 --dport 443 -m conntrack --ctstate NEW -j ACCEPT
root ~
```

- Block all other traffic/websites

```
root ~
# iptables -A FORWARD -j REJECT
```

FORWARD chain linked with CDAC chain.

```
root ~
# iptables -A CDAC -j RETURN
root ~
# iptables -I FORWARD 1 -j CDAC
```

Verification of final IPTables firewall configuration.

```
root ~
❯ iptables -L CDAC --line-numbers
Chain CDAC (1 references)
num target      prot opt source                destination
1  ACCEPT      udp  --  192.168.25.0/24        student-dc.students.acts.in  udp dpt:domain
2  ACCEPT      udp  --  student-dc.students.acts.in 192.168.25.0/24            udp spt:domain
3  ACCEPT      tcp  --  192.168.25.0/24        student-dc.students.acts.in  tcp dpt:domain
4  ACCEPT      tcp  --  student-dc.students.acts.in 192.168.25.0/24            tcp spt:domain
5  ACCEPT      all  --  anywhere                anywhere                    ctstate RELATED,ESTABLISHED
6  ACCEPT      tcp  --  192.168.25.10          a23-217-155-252.deploy.static.akamaitechnologies.com tcp dpt:https ctstate NEW
7  ACCEPT      tcp  --  192.168.25.10          a23-218-230-105.deploy.static.akamaitechnologies.com tcp dpt:https ctstate NEW
8  ACCEPT      tcp  --  192.168.25.10          a2-20-156-84.deploy.static.akamaitechnologies.com tcp dpt:https ctstate NEW
9  ACCEPT      tcp  --  192.168.25.20          server-108-159-80-107.bom78.r.cloudfront.net tcp dpt:https ctstate NEW
10 ACCEPT      tcp  --  192.168.25.20          server-108-159-80-52.bom78.r.cloudfront.net tcp dpt:https ctstate NEW
11 ACCEPT      tcp  --  192.168.25.20          server-108-159-80-26.bom78.r.cloudfront.net tcp dpt:https ctstate NEW
12 ACCEPT      tcp  --  192.168.25.20          server-108-159-80-48.bom78.r.cloudfront.net tcp dpt:https ctstate NEW
13 ACCEPT      tcp  --  192.168.25.20          103.209.97.9                tcp dpt:https ctstate NEW
14 ACCEPT      tcp  --  192.168.25.0/24        142.251.155.119             tcp dpt:https ctstate NEW
15 ACCEPT      tcp  --  192.168.25.0/24        142.251.150.119             tcp dpt:https ctstate NEW
16 ACCEPT      tcp  --  192.168.25.0/24        142.251.157.119             tcp dpt:https ctstate NEW
17 ACCEPT      tcp  --  192.168.25.0/24        142.251.151.119             tcp dpt:https ctstate NEW
18 ACCEPT      tcp  --  192.168.25.0/24        142.251.153.119             tcp dpt:https ctstate NEW
19 ACCEPT      tcp  --  192.168.25.0/24        142.251.156.119             tcp dpt:https ctstate NEW
20 ACCEPT      tcp  --  192.168.25.0/24        142.251.152.119             tcp dpt:https ctstate NEW
21 ACCEPT      tcp  --  192.168.25.0/24        142.251.154.119             tcp dpt:https ctstate NEW
22 ACCEPT      tcp  --  192.168.25.0/24        student-dc.students.acts.in tcp dpt:http ctstate NEW
23 ACCEPT      tcp  --  192.168.25.0/24        student-dc.students.acts.in tcp dpt:https ctstate NEW
24 RETURN      all  --  anywhere                anywhere
```

```
root ~
❯ iptables -L FORWARD --line-numbers
Chain FORWARD (policy ACCEPT)
num target      prot opt source                destination
1  CDAC         all  --  anywhere                anywhere
2  REJECT       all  --  anywhere                anywhere            reject-with icmp-host-prohibited
```

```
root ~
❯ iptables -L
Chain INPUT (policy ACCEPT)
target      prot opt source                destination
ACCEPT      all  --  anywhere                anywhere                    state RELATED,ESTABLISHED
ACCEPT      icmp --  anywhere                anywhere
ACCEPT      all  --  anywhere                anywhere
ACCEPT      tcp  --  anywhere                anywhere                    state NEW tcp dpt:ssh
REJECT      all  --  anywhere                anywhere            reject-with icmp-host-prohibited

Chain FORWARD (policy ACCEPT)
target      prot opt source                destination
CDAC        all  --  anywhere                anywhere
REJECT      all  --  anywhere                anywhere            reject-with icmp-host-prohibited
REJECT      all  --  anywhere                anywhere            reject-with icmp-port-unreachable

Chain OUTPUT (policy ACCEPT)
target      prot opt source                destination

Chain CDAC (1 references)
target      prot opt source                destination
ACCEPT      udp  --  192.168.25.0/24        student-dc.students.acts.in  udp dpt:domain
ACCEPT      udp  --  student-dc.students.acts.in 192.168.25.0/24            udp spt:domain
ACCEPT      tcp  --  192.168.25.0/24        student-dc.students.acts.in  tcp dpt:domain
ACCEPT      tcp  --  student-dc.students.acts.in 192.168.25.0/24            tcp spt:domain
ACCEPT      all  --  anywhere                anywhere                    ctstate RELATED,ESTABLISHED
ACCEPT      tcp  --  192.168.25.10          a23-217-155-252.deploy.static.akamaitechnologies.com tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.10          a23-218-230-105.deploy.static.akamaitechnologies.com tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.10          a2-20-156-84.deploy.static.akamaitechnologies.com tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.20          server-108-159-80-107.bom78.r.cloudfront.net tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.20          server-108-159-80-52.bom78.r.cloudfront.net tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.20          server-108-159-80-26.bom78.r.cloudfront.net tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.20          server-108-159-80-48.bom78.r.cloudfront.net tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.20          103.209.97.9                tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.0/24        142.251.155.119             tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.0/24        142.251.150.119             tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.0/24        142.251.157.119             tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.0/24        142.251.151.119             tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.0/24        142.251.153.119             tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.0/24        142.251.156.119             tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.0/24        142.251.152.119             tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.0/24        142.251.154.119             tcp dpt:https ctstate NEW
ACCEPT      tcp  --  192.168.25.0/24        student-dc.students.acts.in tcp dpt:http ctstate NEW
ACCEPT      tcp  --  192.168.25.0/24        student-dc.students.acts.in tcp dpt:https ctstate NEW
RETURN      all  --  anywhere                anywhere
```